Calculation questions – 4.6 Waves

AQA GCSE Physics (8463) Higher · Paper 2 · 16 questions · 48 marks · answers at the back

Equations for this topic

Equation	Symbols and units	Exam
v = fλ	v (m/s), f (Hz), λ (m)	Learn it
T = 1/f	T (s), f (Hz)	On sheet
s = vt	s (m), v (m/s), t (s)	Learn it
magnification = image height ÷ object height (triple)	no unit	On sheet
Echoes: distance = speed × time ÷ 2		Rule

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

Grade 4–5

1 A sound wave has a frequency of 50 Hz and a wavelength of 6.8 m. Calculate its speed. [2 marks]

Answer: _____ Unit: _____

2 Calculate the period of a wave with frequency 200 Hz. [2 marks]

Answer: _____ Unit: _____

3 15 water waves pass a point in 5.0 s. Calculate the frequency. [1 mark]

Answer: _____ Unit: _____

4 A radio wave travels at 3.0×10^8 m/s with a frequency of 100 MHz. Calculate its wavelength. [3 marks]

Answer: _____ Unit: _____

Grade 6–7

5 Sound travels at 340 m/s. Calculate the frequency of a sound wave with a wavelength of 0.85 m. [3 marks]

Answer: _____ Unit: _____

6 A student shouts at a cliff and hears the echo 1.2 s later. Speed of sound = 340 m/s. Calculate the distance to the cliff. [3 marks]

Answer: _____ Unit: _____

7 On an oscilloscope, one complete wave covers 4 divisions. The time base is 2.5 ms per division. Calculate (a) the period and (b) the frequency. [3 marks]

Answer: _____ Unit: _____

8 (Triple) A lens forms an image 7.0 cm tall of an object 2.0 cm tall. Calculate the magnification. [2 marks]

Answer: _____ Unit: _____

9 Crests on a rope are 0.40 m apart and the period of the wave is 0.25 s. Calculate the wave speed. [3 marks]

Answer: _____ Unit: _____

10 A microwave has a wavelength of 12 cm. Calculate its frequency. Speed of light = 3.0×10^8 m/s. [3 marks]

Answer: _____ Unit: _____

Grade 8–9

11 (Triple) An ultrasound pulse reflects from a tissue boundary and returns 52 µs after it was sent. Speed of ultrasound in tissue = 1500 m/s. Calculate the depth of the boundary in cm. [4 marks]

Answer: _____ Unit: _____

12 (Triple) A ship's sonar pulse returns from the seabed after 1.6 s. Speed of sound in seawater = 1500 m/s. Calculate the depth of the sea. [3 marks]

Answer: _____ Unit: _____

13 A water wave of frequency 2.0 Hz travels from deep water at 0.30 m/s into shallow water, where it travels at 0.20 m/s. (a) Calculate the wavelength in each region. (b) Explain why the frequency does not change. [5 marks]

Answer: _____ Unit: _____

14 (Triple) After an earthquake, P-waves travel at 8.0 km/s and S-waves at 4.5 km/s. At a seismometer the S-waves arrive 70 s after the P-waves. Calculate the distance to the earthquake. [5 marks]

Answer: _____ Unit: _____

15 The Sun is 1.5×10^{11} m from Earth. Calculate how long light takes to reach Earth, in minutes. [3 marks]

Answer: _____ Unit: _____

16 (Triple) A camera forms an image 1.2 mm tall with a magnification of 0.040. Calculate the object height in cm. [3 marks]

Answer: _____ Unit: _____

Answers – 4.6 Waves

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

Q	Working	Answer
1	$v = 50 \times 6.8$	**340 m/s**
2	$T = 1 \div 200$	**0.0050 s**
3	$15 \div 5.0$	**3.0 Hz**
4	$\lambda = 3.0\times10^{8} \div 100\times10^{6}$	**3.0 m**
5	$f = 340 \div 0.85$	**400 Hz**
6	$s = 340 \times 1.2 = 408$ m there and back; $\div 2$	**204 m**
7	$T = 4 \times 2.5$ ms $= 10$ ms; $f = 1 \div 0.010$	**(a) 0.010 s (b) 100 Hz**
8	$7.0 \div 2.0$	**3.5**
9	$f = 1 \div 0.25 = 4.0$ Hz; $v = 4.0 \times 0.40$	**1.6 m/s**
10	$f = 3.0\times10^{8} \div 0.12$	**2.5×10^{9} Hz**
11	$1500 \times 52\times10^{-6} = 0.078$ m; $\div 2$	**3.9 cm**
12	$1500 \times 1.6 \div 2$	**1200 m**
13	$\lambda = v \div f$: $0.30 \div 2.0$ and $0.20 \div 2.0$. Frequency is set by the source; the same number of waves arrive as leave each second	**(a) 0.15 m, 0.10 m**
14	$d/4.5 - d/8.0 = 70 \rightarrow d \times 0.0972 = 70$ (check: 160 s − 90 s = 70 s)	**720 km**
15	$t = 1.5\times10^{11} \div 3.0\times10^{8} = 500$ s	**8.3 minutes**
16	object $= 1.2 \div 0.040 = 30$ mm	**3.0 cm**

Common mistakes. Forgetting to halve echo times (Q6, Q11, Q12); not converting MHz, ms, μs and cm (Q4, Q7, Q10, Q11).